1.3 Sending messages over the internet

In this lesson

You will learn:

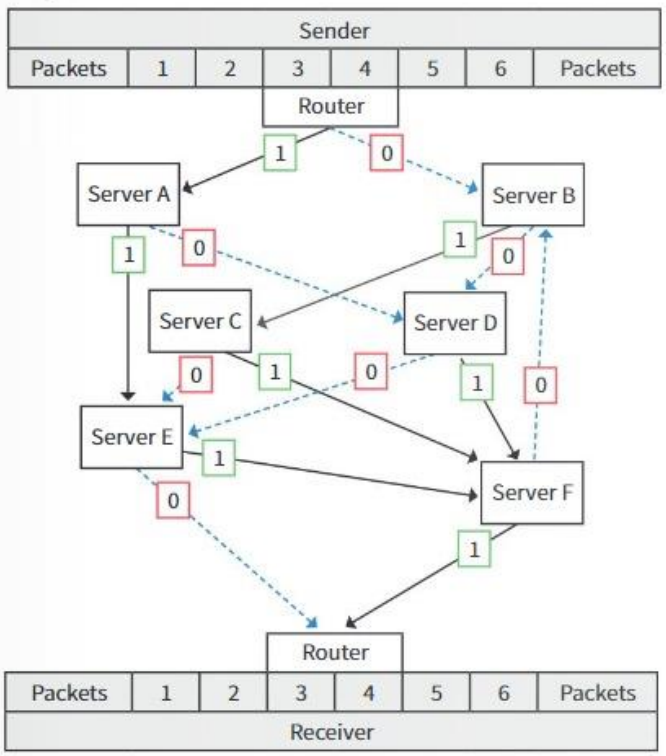
- the rules the internet uses to send messages.

Being connected

Being connected to the internet means you can send and receive messages. There are many types of messages you can send. These include simple text messages, audio, video and photographs.

To a computer, a text message and a video are treated the same – they are both **digital data**. A computer stores all files as digital data made up of zeros and ones. It sends every file across the internet in the same way.

In this lesson you are going to build a model. You will use it to learn how messages are sent over the internet. When you have completed the model, it will look like the one in the diagram. It looks complicated but you will build it step by step before using it in the next lesson.



Activity

Get ready to build your model. You will need a large sheet of paper, two or three coloured pencils or pens, and a ruler. If you don't have a large sheet of paper, stick two sheets of A4 printer paper together along the long edge.

First, add a sender section to the top of your sheet. There are three parts to it:

- the word 'Sender'
- a row labelled 'Packets'. This row should contain six boxes, numbered 1 to 6.
- a box labelled 'Router'.

Add a similar section at the bottom of your sheet. This section is called 'Receiver' and is a mirror image of the 'Sender' section you have drawn at the top of the page.

Do not draw the sections too large. Leave plenty of room in the centre of the sheet.

What are packets?

A computer does not send messages as a single section of data over the internet. Any file that you send over the internet is split into smaller sections before it is sent. The small sections are called **packets**. Whether the message is an email, a photograph or a word-processed file, it is split into small packets before being sent. This makes sending a message easier for the equipment in the network to handle. A packet is about the size of this paragraph (about 550 characters). If you could look inside a packet you would see digital data – 0s and 1s.

When the packets reach their destination, they must be put back together into the original message. You don't have to worry about creating packets or putting them back together. Your computer does it automatically whenever you send a message over the internet.

Many thousands of packets are sent if you are sending a photograph or another large message. Each packet is addressed to the person receiving the message. Each packet is numbered so that the message can be put back together in the right order.

What does a router do?

In Lesson 1.1, you learned that a router is used to connect your school or home network to the internet. Every packet in your message will pass through the router one after the other. When you use the model, you will move packets onto the router box before they are sent.

Activity

Go back to the model you started to draw. In the space in the centre of your page, do the following:

- Draw six boxes. Don't make the boxes too large. Make sure there is plenty of space between them.
- Label the boxes 'Server A' to 'Server F'. Look back at the diagram on page 14 if you're unsure how the boxes should look.

What does a server do?

As discussed in Lesson 1.1, the job of a server is to send files over a network. When you send a message over the internet, it is split into packets. Each packet travels in steps across the internet, jumping from one server to another until it reaches its destination.

Activity

In this activity you will add links, joining servers in your model together. In the real world, each server will be linked to many other servers. You will add just two links from each server in your model. This keep things simple.

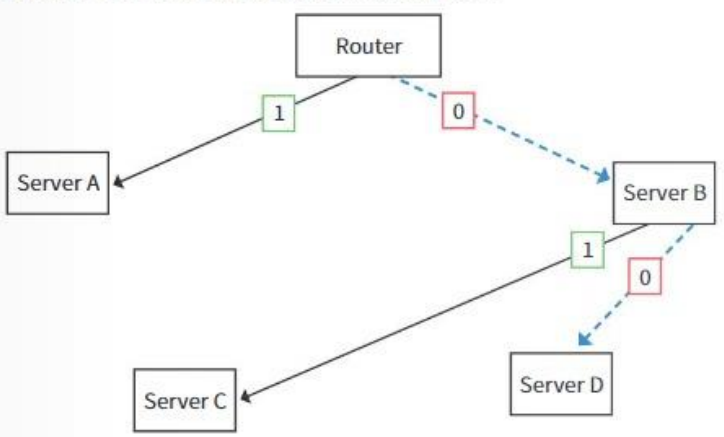
To make your model easy to follow, choose two different line styles.

- Use two different colours.
- Label one style of line '0' and the other '1'.

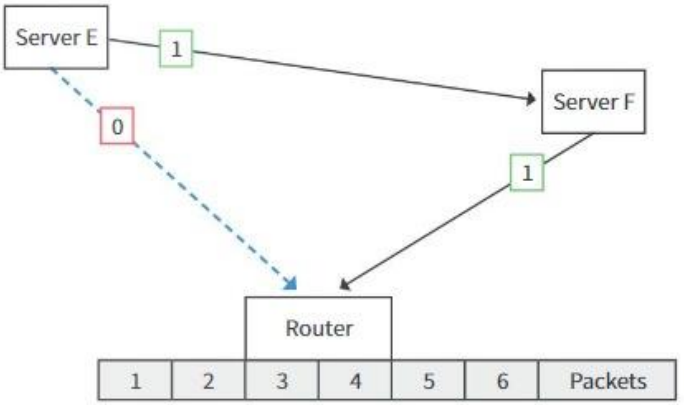
Add an arrow to indicate the direction of the line.

Start from the router at the top of the page. Draw connections to two servers. Use different line styles.

Work through each of the servers, adding two links to the other servers in your model. Make sure each server is linked to at least once.



Make sure two lines (one of each style) link to the router in the receiver, as shown in the image.



Extra challenge

Look back at the completed model from the start of this lesson. Check the model you have drawn to make sure it looks the same. If it doesn't, make sure you understand where you have made mistakes. Update your model.

How a packet travels over the internet

You have learned that a message is passed from server to server as it finds its destination. There is no set route for the message. When a server receives a packet, it must decide where to send it to next. The server will always send a message along the clearest route, avoiding slow or blocked routes.

Test

1 What is a router used for?
2 How does a packet travel across the internet?
3 Why are internet messages sent as packets?
4 Explain why packets sent over the internet have to be numbered.

1.4 Sending messages: a simulation

In this lesson

You will learn:

- how messages are sent over the internet using the packet switching method.

A simulation

In this lesson you will use the model you drew in Lesson 1.3 to show how messages are sent from one computer to another over the internet. The model shows how a message is sent step by step. You are going to use it in a simulation. A simulation copies something that happens in real life.

What you know so far

- All data sent over the internet is digital data.
- A message reaches its destination by being passed from server to server over the internet.
- A message is broken up into small packets before being sent over the internet.
- A router is used to connect a local computer to the internet. All messages sent over the internet will pass through the router.

Preparing for the simulation

You will need:

- your completed model
- a tracking table, which you will create in this lesson
- a coin
- six squares of paper, the same size as the numbered squares in the packets row in your model. Number the squares 1 to 6. These will be your packets of data.

Activity

Create a table like this.

Packet tracker								
	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Step 8
Packet 1								
Packet 2								
Packet 3								
Packet 4								
Packet 5								
Packet 6								

Playing the simulation

Set up your simulation board. Start by putting the six numbered squares in the packets row of the 'Sender' section. The objective of the simulation is to move the message through the network of servers to the receiver section.

You are going to flip the coin during the simulation to decide which path each packet will take between servers. Decide which side of the coin represents '0' and which represents '1'. Make a note so that you remember.

Move Packet 1 to the router. Now you are ready to start.

Rules

- Flip the coin at the start of each round. The result will determine which path all packets will take in that round.
- Move each packet that is in play to the next server using the path (0 or 1) decided by the coin flip. A packet is 'in play' if it is on a server or waiting in the sender's router box. It is easier if you move the packets in number sequence from the lowest to the highest.
- In the table, record the server letter each packet has moved to. When a packet reaches the receiver router, write an 'R' in the table.

Packet tracker								
	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7	Step 8
Packet 1	A	D	E	F	R			
Packet 2	B	D	F	R				
Packet 3	B	C	F					
Packet 4	A	E						
Packet 5	A							
Packet 6								

- Move any pieces that have arrived at the receiver router onto the next available slot in the packets row. Do not put the packets on their matching slot in the receiver's packet row. Put them on the next available slot in the order they arrive. So if Packet 6 arrives fourth in sequence, it goes on slot 4, not slot 6.
- Move the next sender packet onto the router.

Repeat the instructions until all packets have moved from the sender to the receiver. It will take around eight or nine rounds to complete the simulation. There will be up to three or four packets 'in play' at any time.



Activity

Play the simulation, then answer the questions below. You might want to play the simulation in a small group.

Look at the tracking table.

- Did all the packets take the same number of steps to reach the receiver?
- Did all the packets take the same route through the network?
- Did *any* of the packets take the same route through the network?
- Did the packets arrive with the receiver in the same order that they set off from the sender?

It is possible, in fact likely, that the packets you sent arrived out of sequence. There is a final step that needs to be added to the model to make sure that the receiver can understand the message. Do you know what that step is?

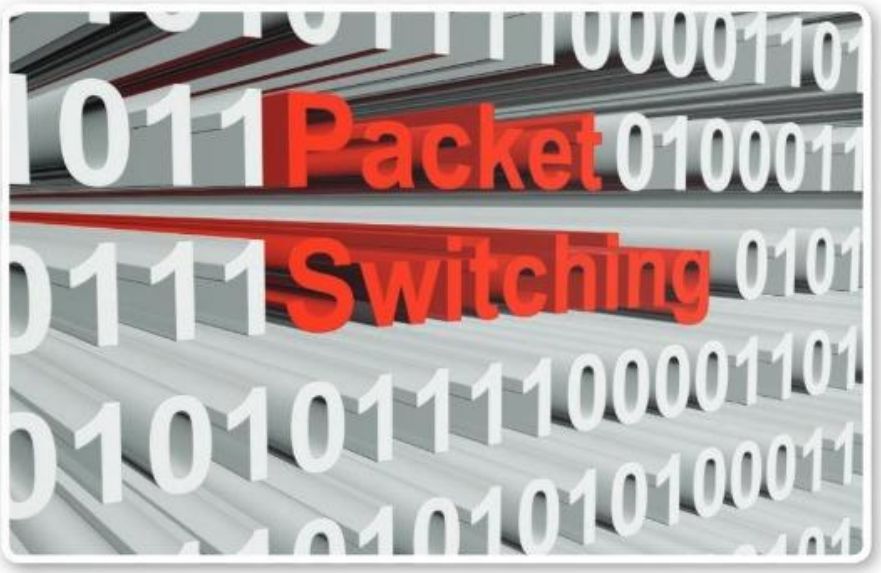
Packet switching

The simulation you have just played is a simple way of showing **packet switching**.

- A message is split into small sections called packets.
- As a packet travels across the internet, it is switched onto the clearest route.

Packet switching is a reliable way of sending messages over the internet. Almost all messages reach their destination correctly. Packet switching is used to send data across local area networks as well as the internet.

Packet switching is carried out automatically by your computer and other hardware in a network. You use packet switching every time you log on to a network.

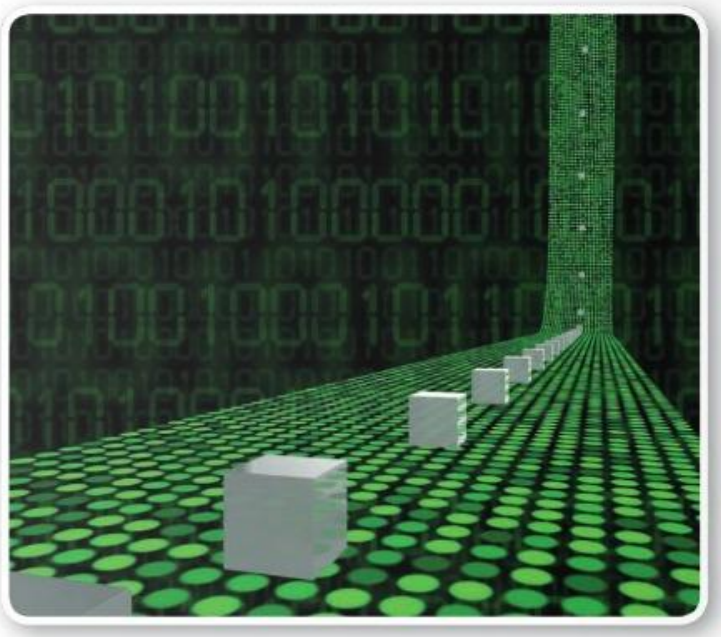


What a packet contains

You already know that a packet contains a small part of a complete message. The data in the packet is stored as digital data.

Each packet contains some extra data.

- The **address** of the person the message is being sent to. When you send an email, you must always add the address of the person you are sending it to (for example, a.friend@gmail.com). When your message is split into packets, each packet must contain the address.
- The **sequence number** of the packet. You saw in the simulation that packets can arrive at their destination in the wrong order. A sequence number allows the receiving computer to sort the packets into the right order.
- The **total number of packets** in the message. The receiving computer uses this information to check if it has received all the packets. If any packets are missing, the computer sends a message back to the sender and asks for them to be re-sent. This is done automatically so you don't have to re-send the whole message.



Extra challenge

Sometimes packets get lost as they are sent across the internet. What problem does that create for the computer receiving a message?

Research the web to find out how packet switching deals with lost packets.

Test

1. What happens if a packet gets lost when being sent across a network?
2. Name three pieces of information that are included in a packet when it is sent over a network.
3. Why can packets arrive at their destination in the wrong order?
4. Describe the term 'packet switching' in your own words.

1.5 Connecting to a network

In this lesson

You will learn:

- how to connect to a network
- basic troubleshooting methods for network connections.



Types of network connection

There are three main ways to connect to a network.

1 **Wired** Your computer is connected to a data point on a wall with a cable. You can only connect to a network if you have a network point nearby. You cannot move your computer. Wired connections are used for desktop computers and sometimes for laptops.

2 **Wireless (Wi-Fi)** A cable isn't needed for a wireless connection. Signals are sent between your computer and a wireless access point (WAP). Lesson 1.1 has more information about WAPs.

Laptop computers, tablets and smartphones connect to networks using Wi-Fi. Using Wi-Fi means you can move around without easily losing your connection.

Wireless networks are sometimes available in public places like restaurants or shops. A WAP in a public place is called a **hotspot**.

3 **Mobile phone** Smartphones can connect to the internet using a mobile phone connection. A connection can be made anywhere there is a phone signal.

How to connect to a network

School network

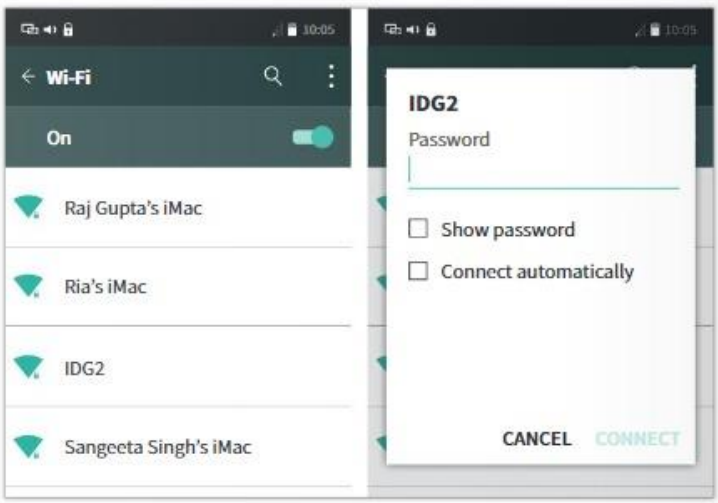
Everyone who is allowed to use the school network is given a username and password. You use the username and password to connect to the network. The login process is the same whether you connect by a wired or wireless connection.



Home network

You do not need a username and password for your home network. Everyone uses the same login details. To log in to the network, you need to know the network name and **network security key** for the router. This information is on a sticker on your router. The network security key works like a password.

When you switch on your computer, you will be shown a list of available networks. Your computer will detect your home network and any other networks close to you. It will detect your neighbour's network, if they are close enough, and any public hotspots.



When you click on the name of your network, you are asked to enter the network security key. Entering it correctly will connect you to the network. When you have entered the key once, your computer will remember it. If you tick the 'Connect automatically' box, you won't need to log on in future. Your computer will automatically connect to the network.

Solving network problems

When you are solving computer problems, you will be working with electrical equipment. Your priority is to stay safe. Always follow your school computer use safety rules. Always check with your parents before trying to deal with problems with your home network.

Stay safe

Always follow these safety rules.

- Switch off power points at the wall before connecting or disconnecting a mains electricity cable.
- Do not open the case of your computer or any other device.
- Never use force when connecting or disconnecting a cable or component.
- Make sure all cables are disconnected before attempting to move a device (for example, a printer).
- Do not lift heavy equipment on your own.
- If in doubt, STOP and ask for help.



Be creative

Create a poster to encourage people to act responsibly and safely when solving computer problems.

Make notes

When you are trying to solve network problems, record:

- the location of the computer
- the steps you have followed
- any error messages you see on-screen.

Username and password

If you are having problems logging on to a network, check you are using the correct username and password.

A password is case-sensitive. That means it matters if you use upper case letters instead of lower case ones. Check you haven't left 'Caps Lock' on.

If you have forgotten your password, use the 'Forgot password?' link to change your password. You will need an email address that a new password can be sent to. At school, you will need to ask your IT technician to change your password.

Wired networks

If you are having network problems with a wired connection, check that the cable connecting your computer to a network point is connected securely at both ends.

The connection socket on your computer will usually have a small green light. If the light is flashing, your computer is connected.

There are two things you can check if the light isn't flashing.

- The cable connecting your computer to the network may be faulty. Changing it for one you know is not faulty will solve the problem.
- The network point you are connected to may be faulty. Try another network point.



Only try to solve problems with your school network if your school IT policy allows it. Report the problem to the IT technician or your teacher and use another computer until the problem is fixed.

Wireless networks

If you cannot make a wireless connection to a network, check if you have wireless enabled on your computer. Find the Wi-Fi icon on the tool bar at the bottom of your screen. If it has a red cross on it, wireless on your computer may be turned off.

The picture on the right shows the wireless network symbol.



Click the Wi-Fi icon to open the wireless control box. You can turn wireless on by clicking the button marked 'Wi-Fi'.

If you are using a tablet or smartphone. you will find the same Wi-Fi icon on your device. If the icon is grey, wireless is turned off. Tap the icon to turn wireless on.

Home network

If you are having problems connecting to your home network, check:

- you are entering your password correctly
- that wireless is turned on
- whether other people in the house have a connection. If they do, the problem is likely to be with your device, not the router.

There are some basic checks you can carry out on your home network router.

- Is the router plugged into an electricity socket? Is the socket switched on?
- Check that any cables connected to the back of your router are secure.

If everything else fails, reset your router. Turn off the power switch or turn off the electricity socket. Wait 30 seconds then turn the router back on again. Do not press any button marked 'reset' on the router.



Activity

Write a list of instructions called 'How to connect to a network'.

Design the instructions for students who are about to start using computers, to help them study at school and at home.

Extra challenge

Work in a small group. Use your multimedia skills to create an audio or video guide to network connectivity. You can use a phone or recorder. Record yourself reading the instructions you wrote, or make a short video of someone connecting to the school network and explaining how to do it.

Test

1. List three ways to connect to a network.
2. What is a hotspot?
3. You are unable to connect your computer to your home network. List four problems that could be preventing you from connecting.
4. You want to connect a laptop to your school network. Why is a wireless connection better than a wired connection?